

### 9.3.2 Initial Procedure (2002)

The first published estimator is based on the ISO 11843-2:2000 standard; it is known as the calibration curve method. The following formulas apply to any specification limit ( $SL$ ) selected by the regulator. The parameter  $k_\alpha = t_{1-\alpha, \nu}$  corresponds to the quantile of Student's  $t$  law for the probability  $\alpha$  and the number of degrees of freedom  $\nu$ . In an equivalent way  $k_\beta = t_{1-\beta, \nu}$  is the quantile for the probability  $\beta$ .

Intermediate quantity

$$IQ = \frac{s_e}{a_1} \times \sqrt{\frac{1}{J} + \frac{1}{IJ} + \frac{(SL - \bar{X})^2}{SCE_X}} \quad (9.2)$$

Decision limit

$$CC\alpha = SL + k_\alpha \times IQ \quad (9.3)$$

Detection capacity (approximation)

$$CC\beta = CC\alpha + k_\beta \times IQ \quad (9.4)$$

The parameter  $SL$  refers to any official compliance specification and takes different names depending on the regulatory text:  $MRL$ , maximum content ( $MC$ ), reference value, and so on. Parameters  $I$  and  $J$  represent the number of calibrators and the number of replicates per calibrator, respectively. These notations differ from those used so far for the accuracy profile.

The so-called “intermediate quantity”  $IQ$  was introduced to facilitate the verification of a worksheet developed to calculate the parameters (see Chapter 10). In addition, it is possible to calculate the previously defined Critical response value denoted  $Y_0$ . It is the smallest response that allows for decision-making; it can be compared to the LOD before it is converted into concentration.

Critical response value

$$Y_0 = a_0 + k_\alpha \times s_e \sqrt{\frac{1}{J} + \frac{1}{IJ} + \frac{\bar{X}^2}{SCE_X}} \quad (9.5)$$

In these formulas,  $a_0$  and  $a_1$  represent the coefficients of the calibration curve (which is assumed to be a simple linear regression model), the intercept or blank, and slope or sensitivity, respectively and  $s_e$  the residual standard deviation (see Section 2.2).

### 9.3.3 Modified Procedure (2021)

In the 2021 regulation update, a new and remarkably interesting calculation procedure was introduced, comparable to the principles of ISO Guide 98-4 on the declaration of conformity when considering the MU; this procedure is described in Section 8.2.

It consists in calculating a rejection zone by combining the regulatory specification limit  $SL$  and the MU of the method obtained by the laboratory at this concentration. The main difference with the ISO 98-4 guide is that the latter introduces the idea of

**Table 9.2** Varied procedures of calculating  $CC\alpha$  and  $CC\beta$ .

| $n^\circ$                                                                        | Principle of the procedure                                   | Equations                                                             |
|----------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------|
| <i>Limit of decision for the purpose of confirmation (<math>CC\alpha</math>)</i> |                                                              |                                                                       |
| Prohibited substances                                                            |                                                              |                                                                       |
| 1                                                                                | Calibration curve method according to ISO 11843-2            | Equations (9.2) and (9.3) with $SL = VR$ or $MC$ and $\alpha = 0.01$  |
| 2                                                                                | 20 blanks and calculation of the signal-to-noise ratio $S/N$ | $CC\alpha = 3.0 \times S/N$                                           |
| 3                                                                                | Measurement uncertainty method                               | $CC\alpha = LCL + k_{\alpha=0.01} u_c(LCL)$ (9.6)                     |
| Authorized substances                                                            |                                                              |                                                                       |
| 4                                                                                | Calibration curve method according to ISO 11843-2            | Equations (9.2) and (9.3) with $SL = MRL$ or $MC$ and $\alpha = 0.05$ |
| 5 <sup>a</sup>                                                                   | Measurement uncertainty method                               | $CC\alpha = LMR + k_{\alpha=0.05} u_c(LMR)$ (9.7)                     |
| 5 <sup>b</sup>                                                                   |                                                              | $CC\alpha = TM + k_{\alpha=0.05} u_c(TM)$ (9.8)                       |
| <i>Detection capability for screening purposes (<math>CC\beta</math>)</i>        |                                                              |                                                                       |
| Prohibited substances                                                            |                                                              |                                                                       |
| 6                                                                                | Calibration curve method according to ISO 11843-2            | Equation (9.4) with $\beta = 0.05$                                    |
| 7                                                                                | Measurement uncertainty method                               | $CC\beta = STC + k_{\beta=0.05} u_c(STC)$ (9.9)                       |
| Authorized substances                                                            |                                                              |                                                                       |
| 8                                                                                | Calibration curve method according to ISO 11843-2            | Equation (9.4) with $\beta = 0.05$                                    |
| 9                                                                                | Measurement uncertainty method                               | $CC\beta = LMR * 1.5 + k_{\beta=0.05} u_c(STC)$ (9.10)                |

Source: Adapted from Commission Implementing Regulation [10].

a *guard band* calculated from the MU, subtracted or added as appropriate to penalize the measurement. The higher the MU, the wider the guard band and the more challenging it is to reach the specification limit.

In this chapter and Table 9.2, it is called the MU method. The nature of the specification limit is the starting point of the calculation. As already mentioned, European regulation recognizes three types of  $SL$ :

|            | Regulatory specification limit | Domain of application                                          |
|------------|--------------------------------|----------------------------------------------------------------|
| <i>MRL</i> | Maximum residue limit          | Many “pharmacologically active” substances are authorized [11] |
| <i>MC</i>  | Maximum content                | Authorized coccidiostats that contaminate other foods[12]      |
| <i>RV</i>  | Reference value                | Prohibited substances defined in [9]                           |

In practice, the regulatory specification limits are established by expert groups summoned by the European Commission. Other non-European expert groups may sometimes set different limits, for

the US-FDA [5]. For a molecule of interest, the specification limit depends on its toxicity, the foodstuff in which it is found, and its consumption level. All these factors condition the potential risks for consumer exposure and will make it possible to establish health standards.

They are supposed to be scientifically based but can sometimes be commercially or health-oriented. The reference value  $RV$ , defined here, is not obtained in the same way as the reference value used for interpreting clinical biology analyses. Section 8.1.2 describes the elaboration methodology for each context: food hygiene and medical diagnosis. In addition to the specification limits, other parameters are involved in the calculation. They are summarized below:

|            | Parameter                      | Definition                                                                                                                                                                                                               |
|------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>STC</i> | Screening target concentration | The concentration at or below the $CC\beta$ at which a screening action classifies the sample as potentially non-compliant and “screen positive” triggers a confirmatory test. It occurs only for prohibited substances. |
| <i>LCL</i> | Lowest calibration level       | Defined by the analyst, this is the lowest concentration to which the measurement system has been calibrated.                                                                                                            |

There is some confusion with these definitions because the official documents do not always give explicit target values. For the analyst, choosing these concentration values is a personal matter. For the screening target concentration (*STC*), it looks like an expert LOD and the lowest calibration level (*LCL*) is experimentally set during calibration.

Table 9.2 is an attempt to present in a condensed manner the various procedures and formulas for calculating  $CC\alpha$  and  $CC\beta$ . A distinction is made between substances that are permitted and those that are not. The numbering reported in the first column of Table 9.2 is not part of the official document. It is added to better identify the diverse calculation procedures. Procedure  $n^{\circ}2$ , although questionable, has been retained for compatibility between the old and new values, but it is intended to disappear in the short term. It is important to emphasize that all procedures allow us to calculate the results directly in the same units as the concentration.

### 9.3.4 Example of Calculation

An example of the application of the “calibration curve method according to ISO 11843-2” is provided by the Resource P worksheet. It is an application of procedures numbered 4 and 8 in Table 9.2. The starting point is to run a set of data used to construct a calibration line which must include  $I > 3$  calibrator levels for which  $J > 2$  replicates. These data occupy cells B3 to C10.

The application of the Excel built-in function `LINEST` allows us to easily obtain the main statistics of the calibration line. In the example, the *MRL* is 200 mg/kg and is shown in cell B11. Finally, the results are  $CC\alpha = 215$  and  $CC\beta = 231$  mg/kg.

A drawback is that these criteria seem to be available only if the calibration model is linear and the data are normally distributed. So, the ISO 11843 series of standards has been expanded over the years to include other parts on various calibration models or probability distributions.

Under the 2021 EU regulation, the coverage factors  $k_1$  and  $k_2$  that appear in various formulas should depend on the number of measures. We are then facing two situations.

- If the number of measurements is important, the quantile of the standardized normal distribution, denoted  $z$ , can be used instead of the Student's  $t$ :

$$k_1 = k_{\alpha=0.01} = k_{\beta=0.01} = z_{(0.99)} = 2.33 \text{ (unilateral at 99\%)}$$

$$k_2 = k_{\alpha=0.05} = k_{\beta=0.05} = z_{(0.95)} = 1.64 \text{ (unilateral at 95\%)}$$

Using the built-in statistical functions of Excel, it gives:

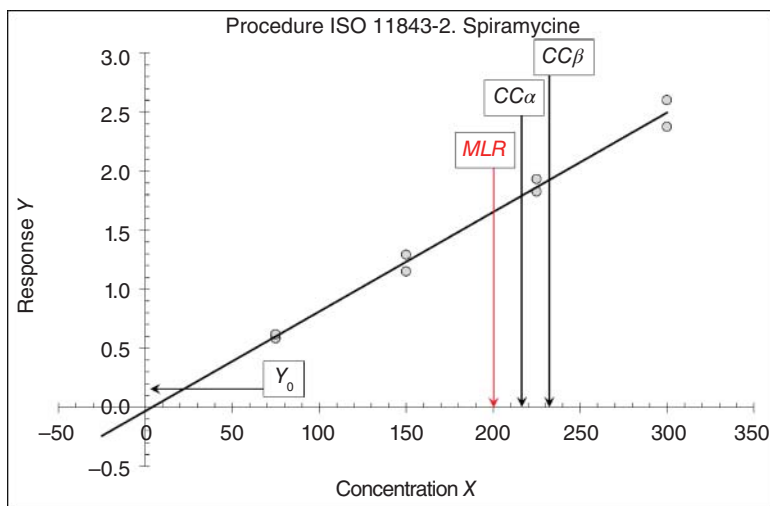
$$k_1 = \text{NORMAL.STANDARD.INVERSE}(0.95)$$

$$k_2 = \text{NORMAL.STANDARD.INVERSE}(0.99)$$

- Otherwise, especially for authorized substances, the Student's  $t$  distribution law can be reasonably applied instead, taking into account the number of results collected during the validation of the method. In this case, the number of degrees of freedom or the number of effective measurements must be used  $N_E$ .

| Resource P Decision limit–calibration curve procedure of ISO 11843-2. |                                                        |                 |                                                             |
|-----------------------------------------------------------------------|--------------------------------------------------------|-----------------|-------------------------------------------------------------|
|                                                                       | A                                                      | B               | C                                                           |
| 1                                                                     | Resource P: Calibration curve procedure of ISO 11843-2 |                 |                                                             |
| 2                                                                     |                                                        | Concentration X | Response Y                                                  |
| 3                                                                     |                                                        | 75              | 0.583                                                       |
| 4                                                                     |                                                        | 75              | 0.617                                                       |
| 5                                                                     |                                                        | 150             | 1.150                                                       |
| 6                                                                     |                                                        | 150             | 1.292                                                       |
| 7                                                                     |                                                        | 225             | 1.933                                                       |
| 8                                                                     |                                                        | 225             | 1.825                                                       |
| 9                                                                     |                                                        | 300             | 2.375                                                       |
| 10                                                                    |                                                        | 300             | 2.600                                                       |
| 11                                                                    |                                                        |                 |                                                             |
| 12                                                                    | Maximum Residue Limit                                  | 200             | mg/kg                                                       |
| 13                                                                    |                                                        |                 |                                                             |
| 14                                                                    | Calibration curve                                      | a1              | a0                                                          |
| 15                                                                    | Coefficients                                           | 0.0084          | -0.0333                                                     |
| 16                                                                    | Standard deviations                                    | 0.0004          | 0.0730                                                      |
| 17                                                                    | Residual std. dev.                                     | 98.9%           | 0.08428                                                     |
| 18                                                                    | Degrees of freedom                                     | 562.52          | 6                                                           |
| 19                                                                    |                                                        | 4.00            | 0.0426                                                      |
| 20                                                                    | Alpha risk                                             | 0.05            |                                                             |
| 21                                                                    | Beta risk                                              | 0.05            |                                                             |
| 22                                                                    | kalpha                                                 | 1.943           | =TINV(2*B20;C18)                                            |
| 23                                                                    | kbeta                                                  | 1.943           | =TINV(2*B21;C18)                                            |
| 24                                                                    | Average X                                              | 187.5           | =AVERAGE(B3:B10)                                            |
| 25                                                                    | SS(X)                                                  | 56250           | =DEVSQ(B3:B10)                                              |
| 26                                                                    | Number of replicates (J)                               | 2               |                                                             |
| 27                                                                    | Number of calibrants (I)                               | 4               |                                                             |
| 28                                                                    | Intermediate quantity                                  | 7.923           | =(C17/B15)*SQRT((1/B26)+(1/(B27*B26)))+(B12-B24)^2/\$B25)   |
| 29                                                                    | Computation of CC alpha and CC Beta                    |                 |                                                             |
| 30                                                                    | Cca                                                    | 215             | =B12+B22*B28                                                |
| 31                                                                    | CCb                                                    | 231             | =B12+2*B23*B28                                              |
| 32                                                                    |                                                        |                 |                                                             |
| 33                                                                    | Critical response                                      | 0.150           | =C15+B22*C17*SQRT((1/B26)+(1/(B27*B26)))+(B24*\$B24/\$B25)) |

Figure 9.3 illustrates the data of example of the Resource P worksheet and the calculation method used since 2002. It shows the relative positions of the MRL,  $CC\alpha$



**Figure 9.3** Detection capacity for the calibration curve method according to ISO 11843-2.

and  $CC\beta$ . As a supplement, the critical response value  $Y_0$ , as calculated in cell B31 of the worksheet, is also plotted on this graph.

To calculate the combined standard uncertainty  $u_c$  of the different criteria  $MRL$ ,  $MC$  and  $STC$ , the European regulation recommends using the intermediate precision, sometimes incorrectly called intra-laboratory reproducibility as already indicated. This recommendation fully agrees with our recommendations developed in the previous chapters about MAP and uncertainty functions.

These new proposals converge on the idea that the MAP, complemented by the calculation of the uncertainty function, preferably in the form of the standard uncertainty, is perfectly suited to estimating the detection capability of methods as introduced by European legislation. In conclusion, the calculation of  $CC\alpha$  and  $CC\beta$  from an accuracy profile is a straightforward solution. No example is supplied here but any application can easily be done when considering examples in previous chapters.

## References

- 1 Raposo, F. and Ibelli-Bianco, C. (2020). Performance parameters for analytical method validation: controversies and discrepancies among numerous guidelines. *Trends in Analytical Chemistry* 129: 115913.
- 2 Currie, L.A. (1999). Detection and quantification limits: origins and historical overview. *Analytica Chimica Acta* 39: 127–134.
- 3 ICH (2022). International Council for Harmonisation of technical requirements for pharmaceuticals for human use (ICH) Guideline Q2(R2) on validation of analytical procedures Step 2b, Amsterdam.